

AMAZON INDIA DASHBOARD

1. Project Overview

This document provides a complete solution structure for an Amazon India sales dashboard in Excel. The workflow covers data preparation, calculated columns, PivotTables, KPI calculations, charts, dashboard layout, and common calculation errors.

Primary objective: convert the raw order-level dataset into an interactive business dashboard that answers:

- What are the total sales, orders, units sold, profit, average order value and profit margin?
- Which categories and products generate the most sales and units?
- How do monthly sales and profit change over time?
- What is the order-status and revenue-loss picture?
- Which payment and fulfillment methods are most frequently used?
- Which states contribute the most sales and profit?

2. Expected Workbook Structure

Sheet	Purpose
Raw_Data	Original Amazon India transaction data; do not overwrite the source.
Clean_Data	Cleaned data plus calculated columns used by PivotTables.
Pivots	Supporting PivotTables for KPI and chart calculations.
Dashboard	Final interactive dashboard with KPI cards, charts and slicers.
Read_Me / Solution	Optional explanation of formulas, assumptions and refresh steps.

3.Data Columns

Common fields used in this solution are:

Field	Use
Order_ID	Unique order identifier; use Count/Distinct Count according to the dataset.
Order_Date / Date	Monthly and yearly trend analysis.
Category	Category-wise sales, profit and units.
Product	Top-product analysis.
Unit_Price_INR	Price-level analysis; not automatically the same as sales.
Quantity / Units	Units sold.
Sales / Revenue / Amount	Primary sales measure when available.
Cost / COGS	Cost of goods used to calculate profit.
Profit	Profit amount if supplied or calculated.
Payment_Method	Payment-method analysis.
Fulfillment_Method	Fulfillment comparison.
Order_Status	Delivered, Shipped, Returned, Cancelled, etc.
State	Geographic analysis.

4. Important Distinction: Unit Price vs Total Sales

A common dashboard error is using Unit_Price_INR as Total Sales. Unit price is the price of one unit. If each row represents an order line, sales normally depend on both unit price and quantity.

row-level sales formula:

=Unit_Price_INR*Quantity

If the dataset already contains a Sales/Amount/Revenue column that represents the transaction value, use that field instead of rebuilding sales.

Therefore, a PivotTable showing:

Sum of Unit_Price_INR = ₹68,213,996.59

with 10,000 Order_ID records should not automatically be labelled Total Sales. It is the sum of the Unit_Price_INR field. Confirm the dataset definition before using it as revenue.

5. Core Calculated Columns

Calculated Column	Excel Formula / Logic	Purpose
Sales	=Unit_Price_INR*Quantity	Transaction-level sales when no sales field exists.
Profit	=Sales-Cost	Profit per row.
Profit Margin %	=IFERROR(Profit/Sales,0)	Profit as a percentage of sales.
AOV	=Total Sales/Total Orders	Average order value.
Return Flag	=IF(Order_Status="Returned",1,0)	Return-rate calculation.
Cancel Flag	=IF(Order_Status="Cancelled",1,0)	Cancellation-rate calculation.
Month	=TEXT(Order_Date,"mmm")	Monthly grouping.
Month Number	=MONTH(Order_Date)	Correct chronological month sorting.
Year	=YEAR(Order_Date)	Year filtering.

6. KPI Calculations

Build the KPI cards from PivotTables or formulas. The following formulas assume Excel Tables and suitable column names; replace names with the exact headers in your file.

KPI	Formula / Method	Meaning
Total Sales	=SUM(Sales)	Total transaction value.
Total Orders	=COUNTA(Order_ID)	Number of order records; use UNIQUE/ROWS for distinct orders if required.
Units Sold	=SUM(Quantity)	Total quantity sold.
Total Profit	=SUM(Profit)	Total profit.
AOV	=Total Sales/Total Orders	Average sales per order.
Profit Margin %	=Total Profit/Total Sales	Overall profitability ratio.
Return Rate %	=Returned Orders/Total Orders	Share of orders returned.
Cancellation Rate %	=Cancelled Orders/Total Orders	Share of orders cancelled.

7. Profit Margin %

Profit Margin % is not the same as Profit and should not be calculated as a raw count or a currency amount.

formula: Profit Margin % = Total Profit ÷ Total Sales

Then format the result as Percentage with 1–2 decimal places.

Example: if Total Sales = ₹100,000 and Total Profit = ₹15,579, then Profit Margin = 15.579%, not ₹15,579. The ₹15,579 value is the profit amount in this example.

8. Category-wise Units Sold

1. Insert a PivotTable from the cleaned dataset.
2. Put Category in Rows.
3. Put Quantity/Units in Values and set the aggregation to Sum.
4. Rename the value field to Units Sold.
5. Sort Largest to Smallest.

Pivot layout:

Rows	Values	Optional Filter
Category	Sum of Quantity	Year / Order Status

9. Category-wise Sales and Profit

6. Rows → Category.
7. Values → Sum of Sales.
8. Values → Sum of Profit.
9. Optional Values → Sum of Quantity.
10. Insert a clustered column or bar PivotChart.

This creates a direct comparison between category revenue and profitability.

10. Top 10 Products by Sales

11. Create a PivotTable.
12. Rows → Product.
13. Values → Sum of Sales.
14. Sort Sales from Largest to Smallest.
15. Apply Value Filters → Top 10 → Top 10 Items by Sum of Sales.
16. Insert a horizontal bar PivotChart.

11. Total Sales

using a PivotTable, place Sales in Values and select Sum. Avoid using Unit_Price_INR alone unless the dataset definition explicitly says that field already represents line revenue.

12. Monthly Sales & Profit Trend

17. Place Order_Date in Rows and group it by Months (and Years if multiple years exist).
18. Place Sales and Profit in Values.
19. Use a Line Chart for the trend.
20. Sort months chronologically, not alphabetically.

If using a Month text column, keep a separate Month Number column and sort Month by Month Number.

13. Order Status Analysis

Visual	Pivot Setup
Order Status Breakdown	Rows → Order_Status; Values → Count of Order_ID
Sales by Status	Rows → Order_Status; Values → Sum of Sales
Return Rate	Returned Orders ÷ Total Orders
Cancellation Rate	Cancelled Orders ÷ Total Orders

14. Payment Method Analysis

21. Rows → Payment_Method.
22. Values → Count of Order_ID or Sum of Sales.
23. Insert a doughnut or column chart depending on the dashboard layout.

For management reporting, show both order volume and sales value where possible.

15. Fulfillment Analysis

24. Rows → Fulfillment_Method.
25. Values → Count of Order_ID and/or Sum of Sales.
26. Use a doughnut chart for order share or a column chart for sales comparison.

16. State-wise Sales

27. Rows → State.
28. Values → Sum of Sales.
29. Sort largest to smallest.
30. For a Top 10 States visual, apply a Top 10 value filter.
31. Horizontal bar chart for readability.

17. Dashboard KPI Cards

Card	Display
Total Sales	₹ amount
Total Orders	Number
Units Sold	Number
Total Profit	₹ amount
AOV	₹ per order
Profit Margin	%
Return Rate	%
Cancellation Rate	%

18. Dashboard Layout

Area	Content
Header	Amazon India Sales Dashboard + date/filter context
Row 1	Total Sales Total Orders Units Sold Total Profit AOV Profit Margin
Row 2	Monthly Sales & Profit Trend Category-wise Sales/Profit
Row 3	Top 10 Products Order Status Payment Method
Row 4	Fulfillment State-wise Sales Category-wise Units

Side / Top Filters	Year Category State Payment Fulfillment Status
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19. Slicers

- Year
- Category
- State
- Payment Method
- Fulfillment Method
- Order Status

After inserting slicers, use Report Connections to connect each slicer to all relevant PivotTables.

20. Conditional Formatting

- Highlight negative profit in a contrasting font/fill.
- Use data bars for category or state sales tables.
- Highlight high return/cancellation categories.
- Use icon sets sparingly for KPI direction.

21. Common Excel Errors and Fixes

Problem	Cause	Fix
Only one value appears in PivotTable	No useful Row/Column field was added.	Add Category/Product/Status to Rows and the measure to Values.
₹68,213,996.59 appears unexpectedly	Sum of Unit_Price_INR is being displayed.	Check the Values field; use Sales/Revenue or Unit Price × Quantity as appropriate.
Count of Order_ID shows 10,000	Count counts records/nonblank IDs.	Use Sum of Sales for revenue; use distinct count only when unique orders are required.
Profit Margin shows ₹ amount	Profit amount was used without dividing by Sales.	Profit Margin = Total Profit / Total Sales and format as %.
Months are out of order	Month names are text.	Sort Month by Month Number or group the actual Date field.
Top 10 is wrong	Count was used instead of sales.	Use Top 10 by Sum of Sales.
Chart does not respond to slicer	PivotTable is not connected.	Slicer → Report Connections → tick required PivotTables.

22. Data Validation Checklist

- Check duplicate Order_IDs and determine whether duplicates are legitimate line items.
- Check blanks in Order_ID, Date, Sales, Quantity, Category and Status.
- Check that Sales and Cost use the same currency and grain.
- Check negative quantities and negative sales.
- Check that dates are true Excel dates.
- Check that numeric fields are stored as numbers, not text.
- Check category spelling and trailing spaces.

23. Final QA Checklist

- All KPI values update when slicers are changed.
- Total Sales is based on the correct revenue field.
- Units Sold uses Sum of Quantity.
- Profit is Sum of Profit.
- Profit Margin is Profit ÷ Sales.
- Top 10 Products uses Sum of Sales.
- Monthly trend is chronologically sorted.
- Return and cancellation percentages use the correct denominator.
- All PivotCharts have meaningful titles and number formats.
- Currency is formatted in INR (₹).
- Percentages use consistent decimal places.
- Dashboard has no broken links, #REF!, #DIV/0!, or #VALUE! errors.

24. Suggested Number Formats

Metric	Format
Sales / Profit / AOV	₹#,##0.00
Orders / Units	#,##0
Profit Margin	0.00%
Return Rate	0.00%
Cancellation Rate	0.00%
Dates	dd-mmm-yyyy

25. Refresh Procedure

32. Paste/replace new raw data in the source table without changing the headers.
33. Refresh the cleaned/calculated data.
34. Data → Refresh All.
35. Refresh every PivotTable.
36. Check slicers and KPI cards.
37. Review the dashboard for errors before sharing.

26. Final Interpretation Guide

The dashboard should be used to answer business questions rather than simply display numbers. For example, high sales with low profit can indicate discounting or high costs; high units with low sales can indicate lower-priced products; high return or cancellation rates can indicate operational or product issues; and strong sales concentration in a small number of states or products can indicate dependence on specific markets.

27. Reference Values from the Working Session

The following value was visible during the working session and is included here as a diagnostic reference, not as an automatic definition of Total Sales:

Observed Pivot Result	Value	Interpretation
Sum of Unit_Price_INR	₹68,213,996.59	Sum of the Unit_Price_INR field; verify whether this is intended to represent revenue.

Count of Order_ID	10,000	Count of nonblank Order_ID records.
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If the dashboard requires actual sales, calculate from the dataset's defined Sales/Amount/Revenue field or from $\text{Unit_Price_INR} \times \text{Quantity}$, depending on the data model.

28. Submission

A complete Sapphire IQ Amazon India dashboard should contain a clean source table, calculated measures, supporting PivotTables, interactive slicers, KPI cards, and charts covering sales, profit, units, products, orders, payment, fulfillment and geography. The most important validation point is to ensure every KPI uses the correct field and aggregation.

END OF COMPLETE SOLUTION